Liangyu Li

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Education

Washington University in St. Louis — **St. Louis, MO**
Doctor of Philosophy in Computational and Data Sciences — *Aug 2024 – May 2029 (Expected)*

Washington University in St. Louis — **St. Louis, MO**
Master of Science in Biomedical Engineering — *Sep 2025 – May 2029 (Expected)*

Georgetown University — **Washington, DC**
Master of Science in Computer Science (GPA: 3.867/4.000) — *Aug 2022 – May 2024*

The Chinese University of Hong Kong — **Shatin, Hong Kong**
Master of Science in Physics — *Aug 2021 – Jul 2022*

Beijing Normal-Hong Kong Baptist University — **Zhuhai, China**
Bachelor of Science (2:1 Honors) in Computer Science and Technology (Major GPA: 3.47/4.00) — *Sep 2017 – Jun 2021*

Relevant Coursework

- Artificial Intelligence
- Machine Learning
- Deep Learning
- Reinforcement Learning
- Computer Vision
- Data Mining
- Natural Language Processing

Research Experience

LLM for Genomic Regulatory Modeling (MPRA-based DNA Expression Prediction) — **Aug 2025 – Present**

- Designed and implemented deep learning pipelines to predict gene expression from **MPRA DNA sequence data**, leveraging large-scale pretrained genomics foundation models.
- Fine-tuned **InstaDeepAI/Nucleotide-Transformer-v2-500M-Multi-Species** using **parameter-efficient LoRA (rank = 32)**; updated only 4.44 % of parameters while achieving test **PCC = 0.6038** on expression prediction.
- Fine-tuned **Borzoi (johahi/borzoi-replicate-0)** for scalar regression from 524 k bp DNA contexts, employing mixed-precision FP16 training, gradient accumulation, and log-space label transformation; achieved test **PCC = 0.5904**.
- Implemented training and evaluation workflows in **PyTorch + Hugging Face Transformers/PEFT**, with reproducible pipelines for tokenization, optimizer scheduling, and early-stopping based on Pearson correlation.
- Demonstrated that **parameter-efficient fine-tuning** of genomic foundation models yields high predictive accuracy on experimental regulatory data, validating generalization across MPRA datasets.

AI for Quantum Sensing in Cells — **Sep 2024 – Dec 2024**

- Developed **deep learning** models to enhance quantum temperature sensing in cells using ESR spectra from nanodiamond systems.
- Compared neural networks with traditional Two-Term Lorentzian fitting methods for spectrum analysis, achieving improved predictive accuracy and robustness.
- Designed **data preprocessing** and augmentation pipelines (e.g., denoising, cubic spline interpolation, PCA-based distribution analysis) to compensate for limited experimental data.
- Applied saliency maps to interpret neural network focus regions, revealing model attention beyond classical peak-centered analysis.
- Conducted experiments on **WT and UCP macrophage cells**, confirming biologically meaningful temperature differences using the trained models.
- Investigated generalization challenges due to distributional shifts across laser powers using K-fold cross-validation and PCA.
- Proposed improvements including symbolic regression, outlier handling, and synthetic data generation for future work.

Inquisitive Conversational Agents by Offline Deep Reinforcement Learning — **May 2023 – May 2024**

- Dataset is the US Supreme Court, created different **measurement methods of Justice's utterance** and considered as the **reward function** of reinforcement learning.
- Using Reward-on-the-Line (ROL) by measuring the **ID and OOD agreement** of action selections to get the **linear relation** to fix rewards.
- Using **Double Conservative Q-Learning (DCQL)** with regularization terms to score the multi-responses to each state, evaluated by P@1.
- The multi-responses generated by **LLM** and using the DCQL model to **score** them, the response with the highest score is the final response.

Physics-Informed Neural Networks for Solving Nonlinear Partial Differential Equations **Sep 2023 – Dec 2023**

- Using **Physics-Informed Neural Networks (PINNs)** to solve different physics nonlinear partial differential equations, including nonlinear Schrödinger equation, Burger's equation, diffusion equation, and Poisson's equation.
- PINNs are trained by incorporating physical laws as soft constraints in the loss function, considering the neural networks as a function.
- Define different initial conditions and boundary conditions in order to **create different loss functions**.

Pun Detection and Location with BERT and Graph Convolutional Neural Networks **Mar 2023 – May 2023**

- **SemEval-2017 Task 7**: each sentence contains a **pun**; need to identify whether a sentence contains a pun or not and locate the punning word.
- Using **Multi-Task Learning**. Using **BERT** to get the embedding of each sentence, building **dependency trees** for each sentence to be the **graphs**, combine them to become the input of **Graph Convolutional Neural Networks (GCNs)**, another input of the model is the **index** of pun in the sentence.
- Built a **control module** that lets the neural network focus on **learning the features** of a token through the input token index by extracting the corresponding token's features slice from the output tensor of BiLSTM, using Sigmoid to identify whether the word is a pun or not.

An Approach Based on Deep Learning for DDoS Detection and Classification **Oct 2022 – Dec 2022**

- Using MLP, CNN, and RNN (GRU, LSTM) to train on the DDoS dataset (CIC-DDoS2019, Kaggle's DDoS) to do **classification**, try to find the performance by using different features, and implementations to identify DDoS attack traffic, including doing **real-time learning**.
- Using **Autoencoders** to do clustering for DDoS data, we found that the features of benign data are more diverse than DDoS.
- Using **transfer learning**, implement DDoS to collect the data every 1 hour, and use new data to train the model to improve the models.

Machine Learning and Deep Learning for Fluids and Crystal Structures **Jan 2022 – May 2022**

- Using CNN, MLP, and RNN to do **classification** for crystal structures, including Liquid, BCC, FCC, HCP, BCCpre, FCCpre, and HCPpre.
- Using **unsupervised learning** such as Autoencoder, t-SNE, and PCA to do **clustering** for crystal structures and fluids.
- Visualizing energy landscapes of fluids by **manifold learning**, based on g(r) and LJ clusters of water molecules (use all pairwise distances).

Machine Learning and Deep Learning for Gas Sensors and Spectrums **Jan 2022 – May 2022**

- Using MLP, LSTM, GRUs, Ridge/Lasso/Elastic Net/Linear regression to **predict** concentrations of different gases of the mixture gas.
- Data are the values of several gas sensors which are simulated by the absorption spectrums of Fourier-Transform Infrared Spectroscopy (FTIR).
- Need to find the best number of filters and their corresponding region of wavenumber, using R^2 Score to evaluate.
- Filters are doing several Gaussian filters of the bandwidth of 50 cm^{-1}, integral these intervals to become several numbers.

Classifying Patients' Medical Records Using Deep Learning and NLP **May 2020 – Dec 2020**

- **Multi-label classification** of various diseases of patients, the dataset is text medical records of **different diagnoses**.
- Implement MLP, CNN, LSTM, BiLSTM, GRU, CRNN, BERT, ALBERT, TextCNN, FastText, Attention, Heterogeneous Graph Attention Network (HAN), etc. with different features of data, to compare the performance.
- Using back-translation, randomly disorder the words based on stop words to do **data augmentation**.

Artificial Intelligence and Computer Vision Four-Wheel Drive Smart Robot **Sep 2020 – Oct 2020**

- Programming: Python, C, OpenCV, TensorFlow, VMware. Hardware: Raspberry Pi 4B mainboard, Driver Board, electronic parts
- **AI** and other functions: voice interactive, voice-activated movement, infrared obstacle avoidance; WIFI video, voice broadcast, mobile control
- **Computer Vision** functions: face/sphere tracking, face/color/QR-code/gesture recognition, tracks black lines, multi-class object recognition

Professional Experience

Washington University in St. Louis — **Aug 2024 – present**
Graduate Research Assistant — *St. Louis, MO*

- I focus on developing new AI algorithms and improving the performance of AI models. Research interests include Large Language Models (LLMs), Computer Vision, Deep Reinforcement Learning, and AI for Science.

Georgetown University — **Aug 2023 – Dec 2023**
Teaching Assistant — *Washington, DC*

- Help to teach COSC-3440 **Deep Reinforcement Learning**.

Geosys Hong Kong Limited — **Jun 2022 – Aug 2022**
Artificial Intelligence Developer (Intern) — *Hong Kong*

- Using the technology of **artificial intelligence**, **deep learning**, **computer vision**, and **image processing** with PyTorch & OpenCV, **object detection** with YOLOv5, **object tracking** with StrongSORT and OSNet, and **image inpainting (GAN)** with LaMa.
- Corresponding tasks included human face detection, Hong Kong license plate detection, wall defect detection, multi-class object detection in the VR3D platform, and surveillance video real-time object detection and tracking. Developed a webpage for real-time surveillance video.

Tisson Regaltec Communications Tech Co., Ltd. — **Jun 2020 – Jul 2020**
Artificial Intelligence Developer (Intern) — *Guangzhou, China*

- The internship is developing software by using **artificial intelligence**, **machine learning**, **deep learning**, and **natural language processing**. Wrote a chat assistant software to assist communication with customers by displaying **prompts** based on conversation content.
- Using PageRank and TextRank to do **information extraction**, and match relevant content in the knowledge base as prompts. Completed various corresponding tasks including **information extraction**, **multi-label text classification**, word embedding, and so on.

Technical Skills

Languages: Python, Java, C, HTML/CSS, JavaScript, SQL, C++
Developer Tools: PyTorch, TensorFlow, NumPy, scikit-learn, SciPy